PARTIAL CIRCUIT MILESTONE — NOT A RELEASE DESIGN

Global nets join sheets. GEN_ and HUB_ segregate control and phase nets. DC_LINK / PGND shared.

36–58.4 V screening range only. No validated current rating. No PCB or routing.

Protected entry still missing; OV chopper and fault memory populated on dedicated pages.

GEN DRV8353S driver and three low-side current channels

File: gen-driver.kicad_sch

GEN 100 V MOSFET bridge — schematic only

File: gen-bridge.kicad_sch

HUB DRV8353S driver and three low-side current channels

File: hub-driver.kicad_sch

HUB 100 V MOSFET bridge — schematic only

File: hub-bridge.kicad_sch

Unfinished safety / control / supply boundaries

File: external-boundaries.kicad_sch

Bus-fed auxiliary and independent safety supply

File: aux-supply.kicad_sch

Autonomous OV chopper comparator and rail / brake monitors

File: protection-comparators.kicad_sch

Release-before-rearm DFF memory

File: fault-latch.kicad_sch

LVC Schmitt fault inputs without series diode offsets

File: fault-interfaces.kicad_sch

Independent brake chopper power stage

File: brake-chopper.kicad_sch

STM32G474VET6 LQFP100: power, reset, debug and control

File: mcu-core.kicad_sch

MCU 3V3 decoupling — external supervised rail

File: mcu-power.kicad_sch

SW CSA ADC input filters and brake status sense

File: mcu-analog.kicad_sch

3V3 Hall interfaces and optional quadrature encoder

File: mcu-sensors.kicad_sch

GEN independent raw CSA window trips

File: gen-fast-oc.kicad_sch

HUB independent raw CSA window trips

File: hub-fast-oc.kicad_sch

Ratiometric OC window and wired sink fault

File: fast-oc-reference.kicad_sch

Bounded bench DC entry — manual dead-circuit sequencing only

File: dc-entry.kicad_sch

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheets: /

File: powerboard-reva.kicad_sch

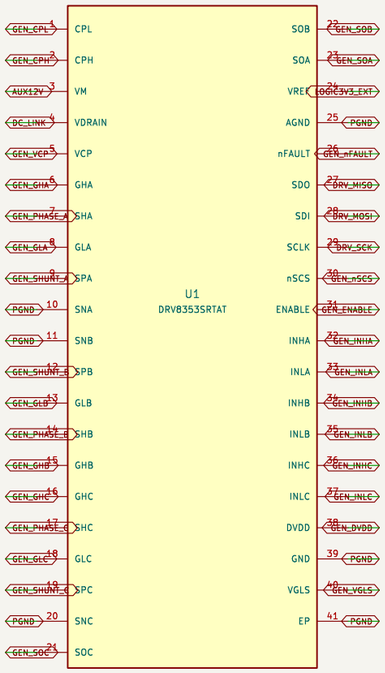
Title: Pedal-by-wire: hierarchical engineering review

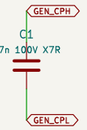
Size: A1	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 1/23

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

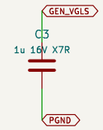
VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

RELEASE BLOCKERS: firmware, OC timing/coverage and transient/thermal qualification unfinished.



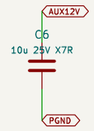


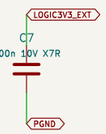


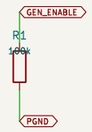


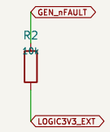


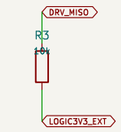


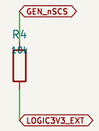












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

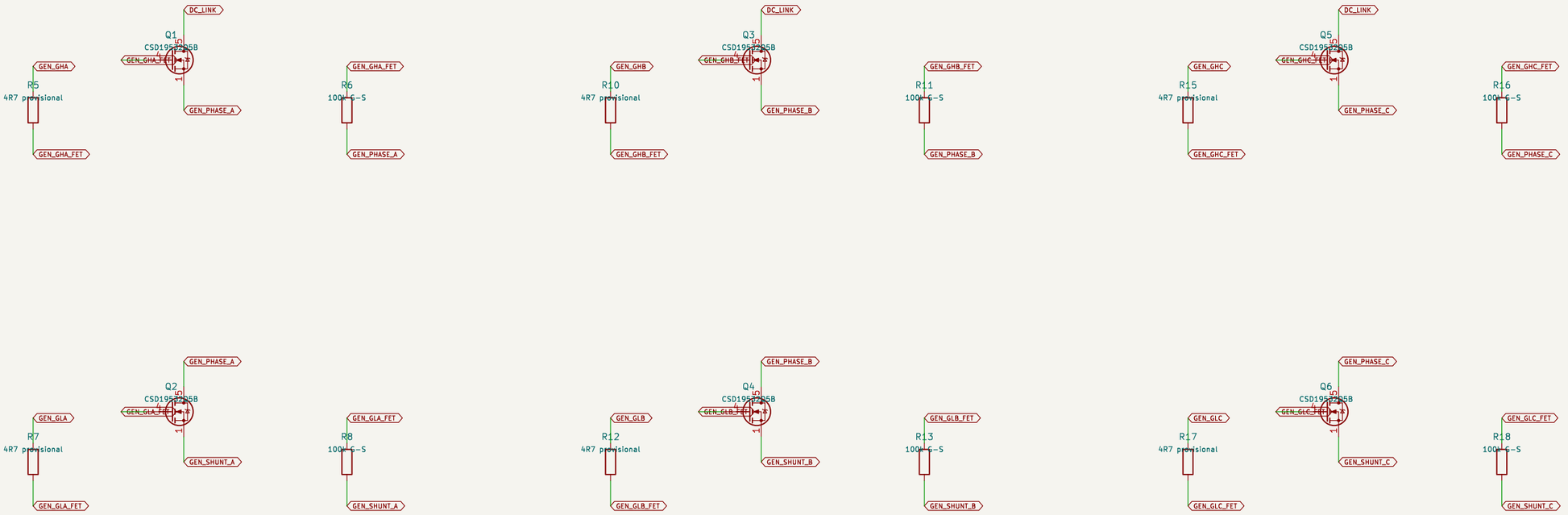
Sheet: /GEN DRV8353S driver and three low-side current channels/
File: gen-driver.kicad_sch

Title: GEN DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 2/23

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /GEN 100 V MOSFET bridge — schematic only/
File: gen-bridge.kicad_sch

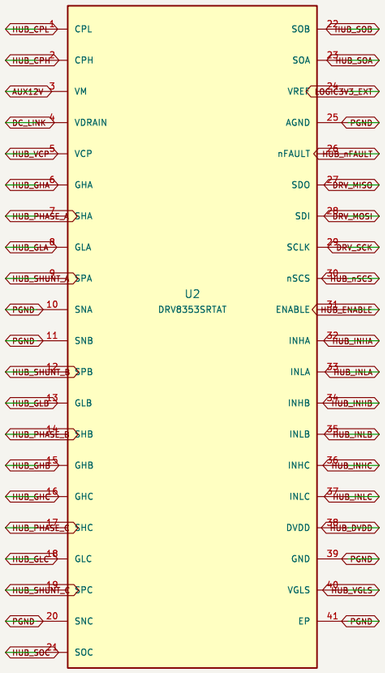
Title: GEN 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 3/23

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

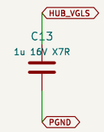
VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

RELEASE BLOCKERS: firmware, OC timing/coverage and transient/thermal qualification unfinished.



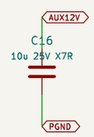


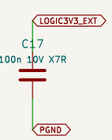




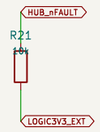


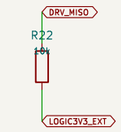


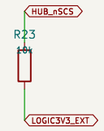












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

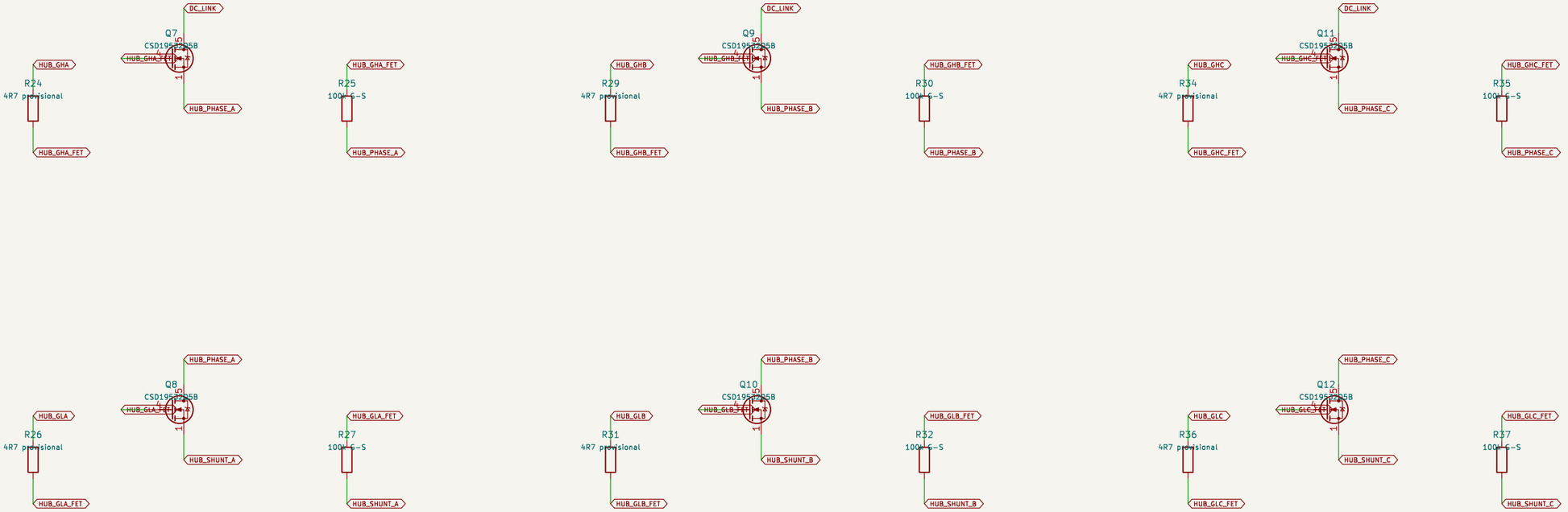
Sheet: /HUB DRV8353S driver and three low-side current channels/
File: hub-driver.kicad_sch

Title: HUB DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 4/23

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /HUB 100 V MOSFET bridge — schematic only/
File: hub-bridge.kicad_sch

Title: HUB 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 5/23

OFFBOARD interfaces: 3V3 supply, brake loop and protected bus entry.

DO NOT ENERGIZE. Protected entry unfinished; MCU and protection remain unqualified.

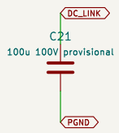












Missing: source fuse, reverse blocking, precharge, DC contactor, bus current sensor and discharge.

Populated: OV chopper, fault latch, brake loop, watchdog, aux supply. Added: raw CSA window trip; timing/coverage unqualified.

MCU and sensor interfaces populated; firmware, thermal sensors, storage/BMS acceptance missing.

PGND is common, not galvanically isolated. Supply sequencing and back-power paths unqualified.

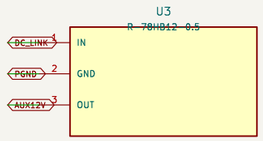
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

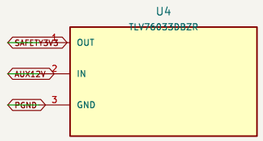
Sheet: /Unfinished safety / control / supply boundaries/
File: external-boundaries.kicad_sch

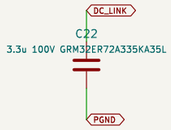
Title: Unfinished safety / control / supply boundaries

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 6/23

R-78HB12-0.5: 17–72V operating, 75V absolute. Entry transient protection still REQUIRED.



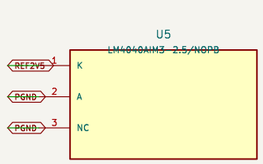




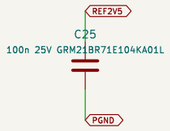












Safety supply is NOT the MCU supply. LOGIC3V3_EXT remains a supervised, external MCU rail.

Supply total/current/thermal budget provisional; no galvanic isolation or stored-energy ride-through.

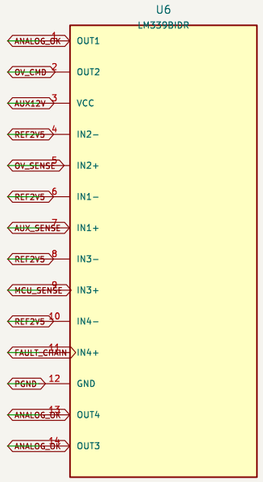
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

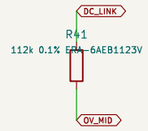
Sheet: /Bus-fed auxiliary and independent safety supply/
File: aux-supply.kicad_sch

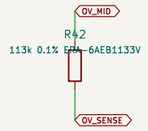
Title: Bus-fed auxiliary and independent safety supply

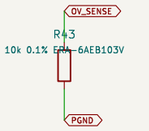
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 7/23

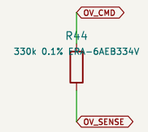
Comparator powered from AUX12V; open collectors pulled to SAFETY3V3. No MCU in OV path.



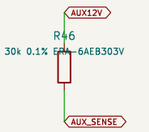


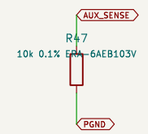


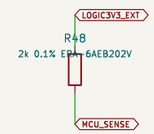


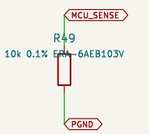


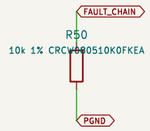




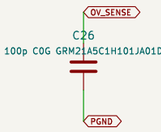














FAULT_CHAIN must close to SAFETY3V3 through normally-closed E-stop + dump thermal switches (NOT rider levers). Open wire trips.

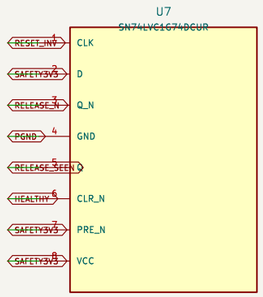
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

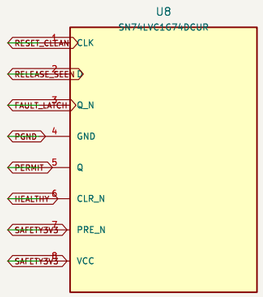
Sheet: /Autonomous OV chopper comparator and rail / brake monitors/
File: protection-comparators.kicad_sch

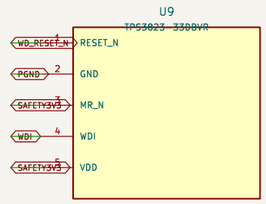
Title: Autonomous OV chopper comparator and rail / brake monitors

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 8/23

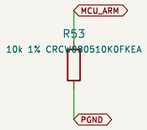
UNSAFE DRAFT: healthy release edge THEN press edge required. No automatic reset.













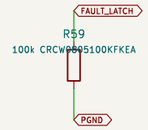


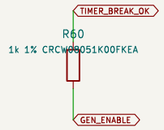


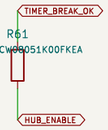


















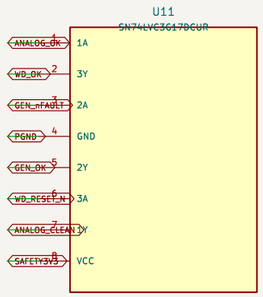
TPS3823 non-A: falling WDI only; hold low during RESET. Stuck RESET stays disabled.

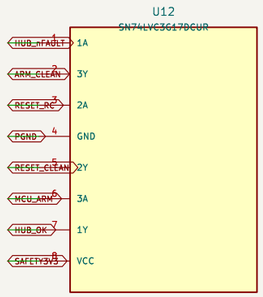
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

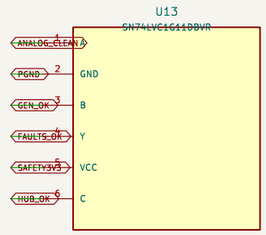
Sheet: /Release-before-rearm DFF memory/
File: fault-latch.kicad_sch

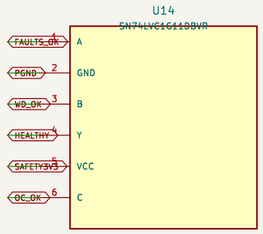
Title: Release-before-rearm DFF memory

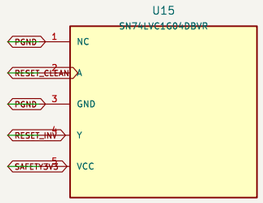
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 9/23

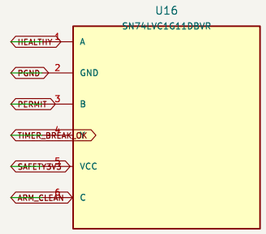
























Default OFF via async supervisor clear. Analog ramp/metastability NOT validated.

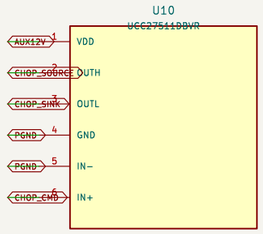
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

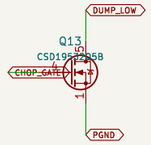
Sheet: /LVC Schmitt fault inputs without series diode offsets/
File: fault-interfaces.kicad_sch

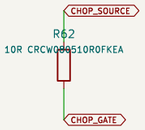
Title: LVC Schmitt fault inputs without series diode offsets

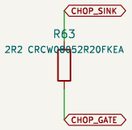
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 10/23

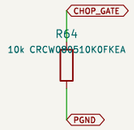
Autonomous OV path ignores bridge ENABLE. External resistor is required across BRAKE_DUMP_RESISTOR connector pins.

















Candidate: 2 x TE HSC3004R7J in series (9.4 ohm). Chassis/heatsink + NC thermal cutoff required.

No resistor cable clamp / fuse / SOA qualification yet. Loss of AUX supply loses bus sink: NOT FAIL SAFE.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

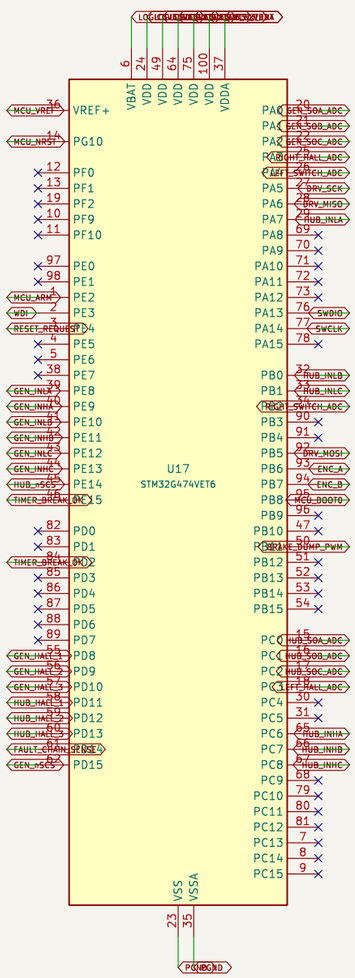
Sheet: /Independent brake chopper power stage/
File: brake-chopper.kicad_sch

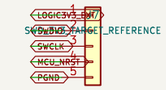
Title: Independent brake chopper power stage

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 11/23

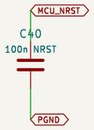
HSI16 internal clock / PLL option: no HSE or LSE fitted; PF0/PF1 and PC14/15 explicit NC.

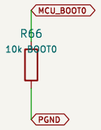
BOOT0 low; keep NRST function and configure boot option bytes before use. No firmware supplied.











BKIN active LOW required in TIM1/TIM8. Separate hardware ENABLE path remains intact.

NC GPIO: leave analog mode; HSI accuracy / PLL / ADC timing require validation.

MCU_ARM / RESET_REQUEST 10k and WDI 1k default-low resistors are on fault-latch page.

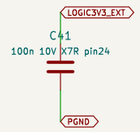
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

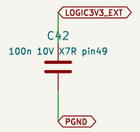
Sheet: /STM32G474VET6 LQFP100: power, reset, debug and control/
File: mcu-core.kicad_sch

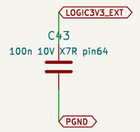
Title: STM32G474VET6 LQFP100: power, reset, debug and control

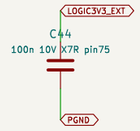
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 12/23

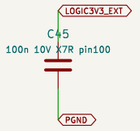
One 100n per VDD pin (24,49,64,75,100) plus VBAT pin6. Place each at pin / ground return.

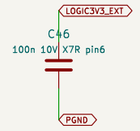


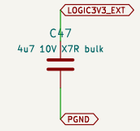


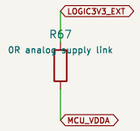


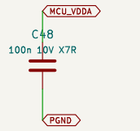


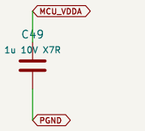


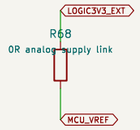


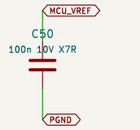


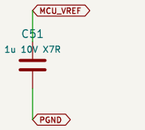












VREFBUF MUST remain disabled: external 3V3 reference. VDDA/VREF tied via OR; not precision isolated.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

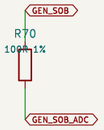
Sheet: /MCU 3V3 decoupling — external supervised rail/
File: mcu-power.kicad_sch

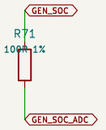
Title: MCU 3V3 decoupling — external supervised rail

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 13/23

100R / 1nF candidate filters: settling / sample aperture / CSA gain remain unqualified.

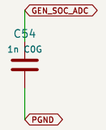




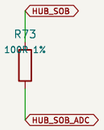


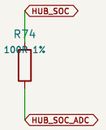




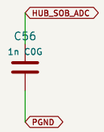


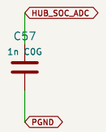


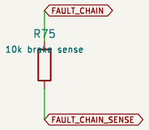


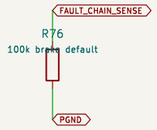


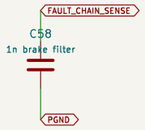












FAULT_CHAIN_SENSE is status ONLY. E-stop / thermal comparator and fault path do not depend on firmware.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /Six CSA ADC input filters and brake status sense/
File: mcu-analog.kicad_sch

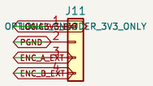
Title: Six CSA ADC input filters and brake status sense

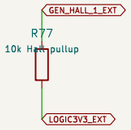
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 14/23

3V3 ONLY: open-collector Hall / encoder or compatible 3V3 outputs. NO 5V tolerance claim.

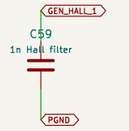


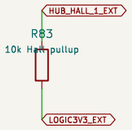






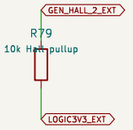












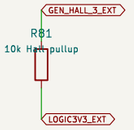






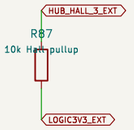






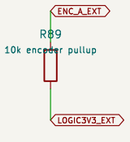




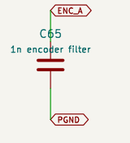


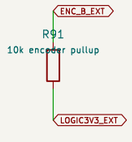


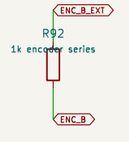


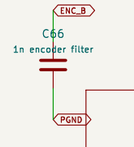












Encoder PB6/PB7: TIM4_CH1/CH2 AF2 DS12288r5 T13 p74. Cable ESD/EMC protection unresolved.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

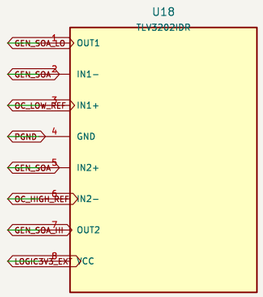
Sheet: /3V3 Hall interfaces and optional quadrature encoder/
File: mcu-sensors.kicad_sch

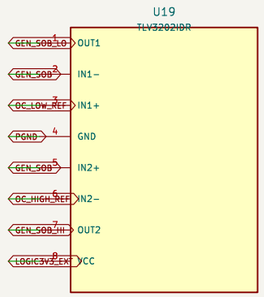
Title: 3V3 Hall interfaces and optional quadrature encoder

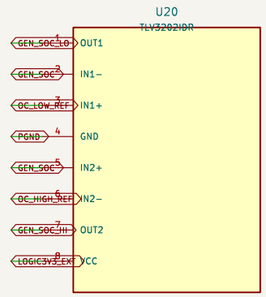
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 15/23

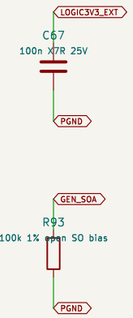
TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

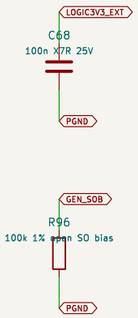
No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.

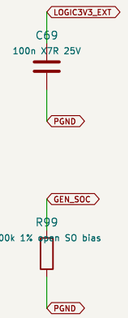


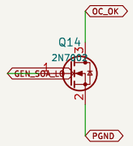


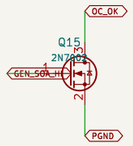


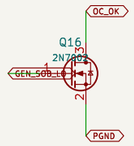


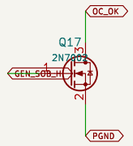


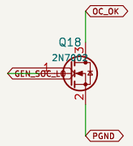


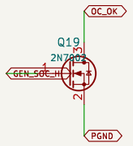


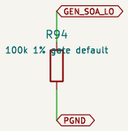


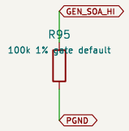


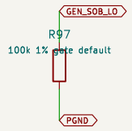


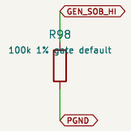


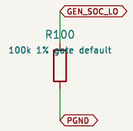


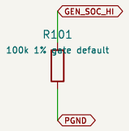












Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

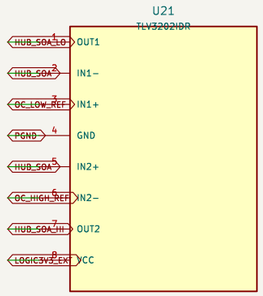
Sheet: /GEN independent raw CSA window trips/
File: gen−fast−oc.kicad_sch

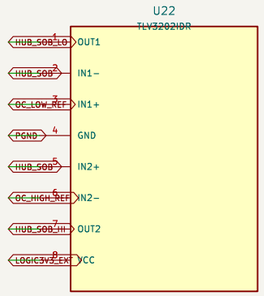
Title: GEN independent raw CSA window trips

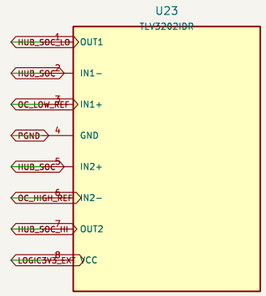
Size: A3	Date:	**Rev: A−PARTIAL**
KiCad E.D.A. 9.0.9		Id: 16/23

TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.

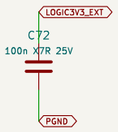




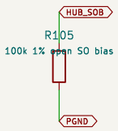




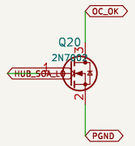


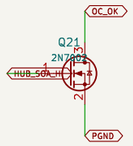


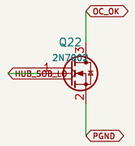


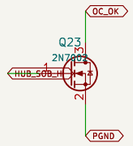


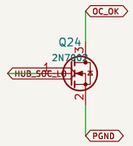


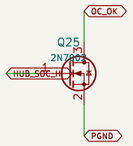


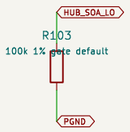


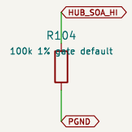


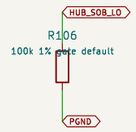


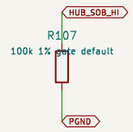


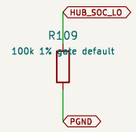


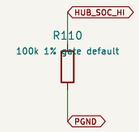












Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE − SAFETY CIRCUITS UNFINISHED

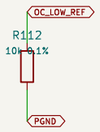
Sheet: /HUB independent raw CSA window trips/
File: hub−fast−oc.kicad_sch

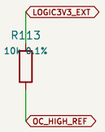
Title: HUB independent raw CSA window trips

Size: A3	Date:	**Rev: A−PARTIAL**
KiCad E.D.A. 9.0.9		Id: 17/23

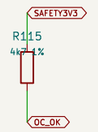
VREF tracks actual driver LOGIC3V3_EXT: 0.852713 / 2.447287 V nominal at 3.3 V.











OC_OK directly enters U14.6, clears U7/U8 and U16: both ENABLE and TIM1/TIM8 break.

No external hysteresis or blanking: first crossing latches. Noise/settling and sink delay unqualified.

Nominal trip +/−19.932 A; provisional measurement +/−30 A. Commissioning cap 10 A, NOT release.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

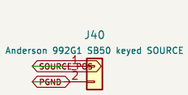
Sheet: /Ratiometric OC window and wired sink fault/
File: fast-oc-reference.kicad_sch

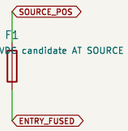
Title: Ratiometric OC window and wired sink fault

Size: A3	Date:	**Rev: A-PARTIAL**
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NOT SAFE TO ENERGIZE integrated drive. No automatic sequencing or reverse-blocking guarantee.

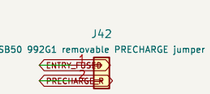
36-58.4V bench interface, 15A TARGET ONLY. Disconnect PM machines before source removal.

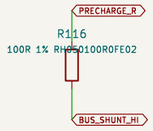


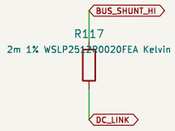


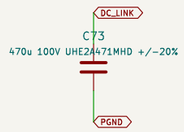


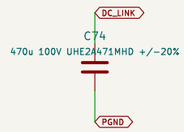


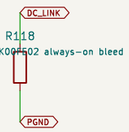














NO approved jumper sequence. Never manipulate J40/J41/J42 energized or bus charged.

J43 raw high common mode: NOT MCU ADC. D20/fuse reverse-fault coordination UNQUALIFIED.

Sink/chopper stays DC_LINK side. MAIN open does NOT make bus safe: PM backfeeds via diodes.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

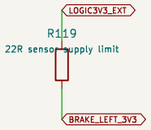
Sheet: /Bounded bench DC entry – manual dead-circuit sequencing only/
File: dc-entry.kicad_sch

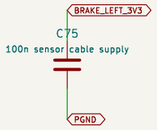
Title: Bounded bench DC entry – manual dead-circuit sequencing only

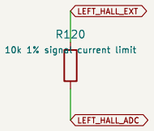
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 19/23

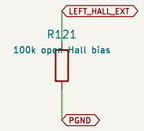
Candidate only: reversible non-load-bearing lever sensors. NO mechanical brake cable alteration.

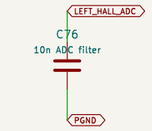




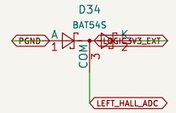


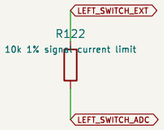


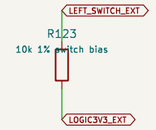


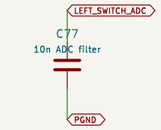




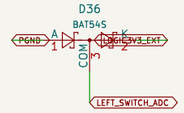












External switch harness: 4k7 series + (10k parallel NC switch). Release=1.055V; press=1.964V.

Open=3V3 / short=0V faults via ADC; firmware NOT implemented. Clamp injection/power-off NOT qualified.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /LEFT rider Hall and supervised binary interface/
File: brake-intent-left.kicad_sch

Title: LEFT rider Hall and supervised binary interface

Size: A3	Date:	**Rev: A-PARTIAL**
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Candidate only: reversible non-load-bearing lever sensors. NO mechanical brake cable alteration.

J45
BRAKE_RIGHT_3V3 ONLY
PGND
RIGHT_HALL_EXT
RIGHT_SWITCH_EXT

R124
22R sensor supply limit
LOGIC3V3_EXT
BRAKE_RIGHT_3V3

C78
100n sensor cable supply
BRAKE_RIGHT_3V3
PGND

R125
10k 1% signal current limit
RIGHT_HALL_EXT
RIGHT_HALL_ADC

R126
100k open-Hall bias
RIGHT_HALL_EXT
PGND

C79
10n ADC filter
RIGHT_HALL_ADC
PGND

D31
PESD3V3U1UL
RIGHT_HALL_EXT
PGND

D35
BAT54S
PGND
LOGIC3V3_EXT
COM
RIGHT_HALL_ADC

R127
10k 1% signal current limit
RIGHT_SWITCH_EXT
RIGHT_SWITCH_ADC

R128
10k 1% switch bias
RIGHT_SWITCH_EXT
LOGIC3V3_EXT

C80
10n ADC filter
RIGHT_SWITCH_ADC
PGND

D33
PESD3V3U1UL
RIGHT_SWITCH_EXT
PGND

D37
BAT54S
PGND
LOGIC3V3_EXT
COM
RIGHT_SWITCH_ADC

External switch harness: 4k7 series + (10k parallel NC switch). Release=1.055V; press=1.964V.

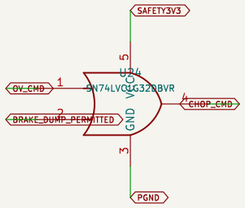
Open=3V3 / short=0V faults via ADC; firmware NOT implemented. Clamp injection/power-off NOT qualified.

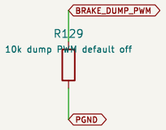
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /RIGHT rider Hall and supervised binary interface/
File: brake-intent-right.kicad_sch

Title: RIGHT rider Hall and supervised binary interface

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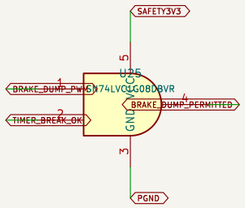


OV always overrides MCU PWM. Normal PWM must require valid rider request AND cool dump permission.

PWM AND TIMER_BREAK_OK: watchdog / E-stop / thermal fault kill normal dump. OV bypass retained.

No torque-sign hardware guard. Firmware must disarm on sensor fault; watchdog / OC / E-stop remain hardware kills.

Firmware entirely unimplemented. Do not energize; brake command control only a design contract.

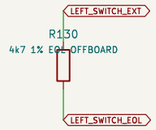


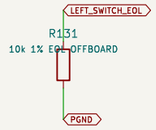


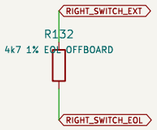
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

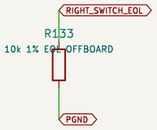
Sheet: /Rider dump PWM OR independent emergency overvoltage/ File: brake-dump-command.kicad_sch		
Title: Rider dump PWM OR independent emergency overvoltage		
Size: A3	Date:	**Rev: A-PARTIAL**
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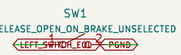
OFFBOARD parts must live at lever, not board. NC contact opens on rider braking. No brake cable modification.

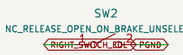












Switch symbol drawn open (braking). Model cables unplugged by removing BOTH EOL resistors and switch.

Welded contact or resistor short can imitate release; Hall plausibility is firmware only, not independent safety.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /OFFBOARD supervised switch end-of-line network — candidate/
File: brake-switch-harness.kicad_sch

Title: OFFBOARD supervised switch end-of-line network — candidate

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